

Question number	Answer	Mark
2(a)(i)	• concentration of sodium hydroxide solution	(1)

Question number	Answer	Mark
2(a)(ii)	An answer that includes any one of the following pairs: • temperature of solution (1) • carry out in a thermostatically controlled water bath (1) or • length of time in solution (1) • start them all at same time / stopwatch (1)	(2)

Question number	Answer	Mark
2(a)(iii)	An answer that includes any five of the following points: • a source material variable taken into account, e.g. length, width, age, mass, hydration level, part of plant extracted from (1) • environmental variable controlled, e.g. temperature, humidity (1) • named procedural variable controlled, e.g. size of masses used (1) • idea of adding masses until fibre breaks / measure the mass {that breaks the fibre / that the fibre can hold before breaking} (1) • repeat and find the {mean / average} (1) • reference to safety procedure (1)	(5)

Question number	Answer	Mark
2(b)(i)	An answer that includes the following points: • for {Baobab, Okra and Kenaf / three of the plants} concentration should be 10% (1) • for Roselle it should be 25% (1)	(2)

Question number	Answer	Mark																	
2(b)(ii)	A table, drawn showing: • suitable table drawn (1) • headings of sodium hydroxide concentration with units, tensile strength with units and the two species (1) • data correctly entered into it (1) Example of table drawn: <table border="1"> <tr> <th rowspan="2">Sodium hydroxide concentration/%</th><th colspan="2">Tensile strength/a.u.</th></tr> <tr> <th>Roselle</th><th>Baobab</th></tr> <tr> <td>10</td><td>0.40</td><td>1.00</td></tr> <tr> <td>15</td><td>0.30</td><td>0.63</td></tr> <tr> <td>20</td><td>0.59</td><td>0.65</td></tr> <tr> <td>25</td><td>0.65</td><td>0.60</td></tr> </table>	Sodium hydroxide concentration/%	Tensile strength/a.u.		Roselle	Baobab	10	0.40	1.00	15	0.30	0.63	20	0.59	0.65	25	0.65	0.60	(3)
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2(c)(i)	An answer that includes the following comparative points: • both {peak at 8% (sodium hydroxide) / go down above 8% (sodium hydroxide) / rise at 25% (sodium hydroxide)} (1) • 0 °C peaks at 30% (sodium hydroxide) but at 20 °C {reaches a plateau / flattens off at 30% (sodium hydroxide)} (1)	(2)

Question number	Answer	Additional guidance	Mark
2(c)(ii)	A description that includes the following points: • an increase in temperature leads to a reduction in degree of swelling (1) • this effect is not linear (1)	Accept reverse argument	(2)